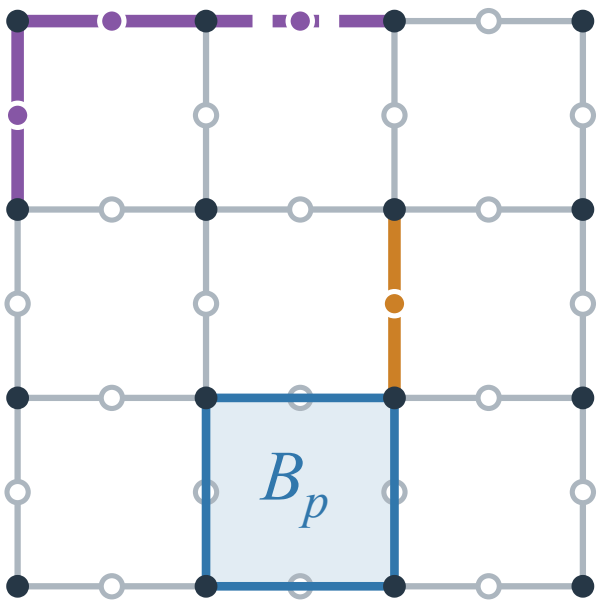
(a) Original lattice

24 link qubits



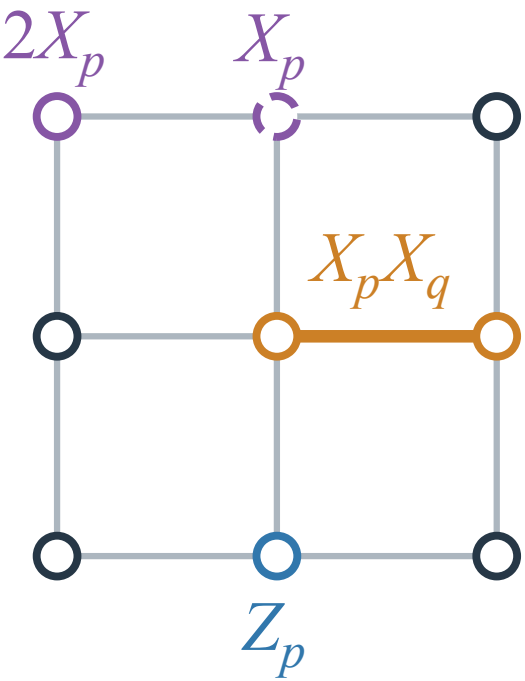
$$\dim \mathcal{H}_{\text{link}} = 2^{24}$$

$$G_v = +1$$

15 independent constraints

(b) Dual lattice

9 plaquette qubits



$$\dim \mathcal{H}_{\text{phys}} = 2^{24-15} = 2^9$$

Plaquette

$B_p \mapsto Z_p$

Interior link

$X_\ell \mapsto X_pX_q$

Boundary link

$X_\ell \mapsto X_p$

Two boundary links at a corner give $2X_p$.